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Centre Number

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Student Number

2023

Year 12

**Mathematics
Extension 2**

Task 3

Validation Task

26/03/2023

Q	Marks
1	/14
2	/16
3	/12
Total	/42

General

Instructions:

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations

Total Marks:

42

This question paper must not be removed from the examination room.

This assessment task constitutes 25% of the course.

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Question 1 (14 marks)

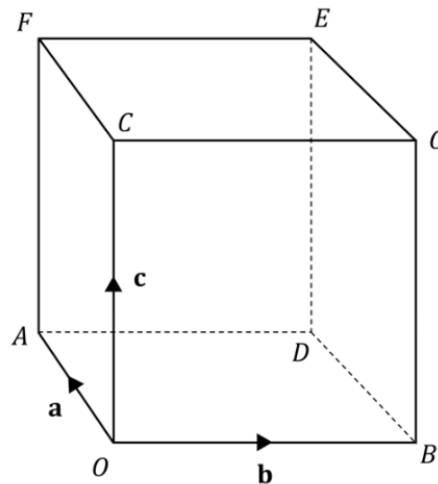
- (a) In the triangle ABC , $\overrightarrow{AB} = -2i + 3j - 4k$ and $\overrightarrow{AC} = -5i - 4j - 7k$. 3

Show that $\angle BAC = 81.9^\circ$ to one decimal place.

- (b) Can the numbers $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ be direction cosines of a line? Give relevant working. 2

- (c) The coordinates of points A and B are $(-2, 5, -7)$ and $(-7, k, 3)$ respectively. Given that the distance from A to B is $5\sqrt{14}$, find possible values of k . 2

- (d) The diagram below shows a cube whose vertices are O, A, B, C, D, E, F and G .



$\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ respectively. A has coordinates $(5, 0, 0)$.

- (i) Express vector $\overrightarrow{OF}$ in component form. 2

- (ii) Explain why $\overrightarrow{FE} = \mathbf{b}$. 1

- (iii) Hence, find the exact distance OE . 2

(e) Find the vector perpendicular to both lines given below:

2

$$\begin{aligned}x + 1 &= \frac{y + 1}{-3} = \frac{z + 1}{2} \\ \frac{x - 3}{2} &= \frac{y - 5}{3} = z - 7\end{aligned}$$

End of Question 1

Question 2 (16 marks)

- (a) Relative to the fixed origin O , the following vector equations are given.

$$\overrightarrow{OA} = \begin{pmatrix} 0 \\ 8 \\ 3 \end{pmatrix} \text{ and } \overrightarrow{OB} = \begin{pmatrix} 1 \\ 13 \\ 1 \end{pmatrix}$$

- (i) Find the vector equation for the straight line l_1 which passes through A and B . 1

- (ii) The straight line l_2 has vector equation: 2

$$\vec{r} = \begin{pmatrix} 7 \\ 0 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$$

Where μ is a scalar parameter.

Show that l_1 and l_2 do not intersect.

- (iii) Find the position vector of C , given that it lies on l_2 and $\angle ABC = 90^\circ$. 2

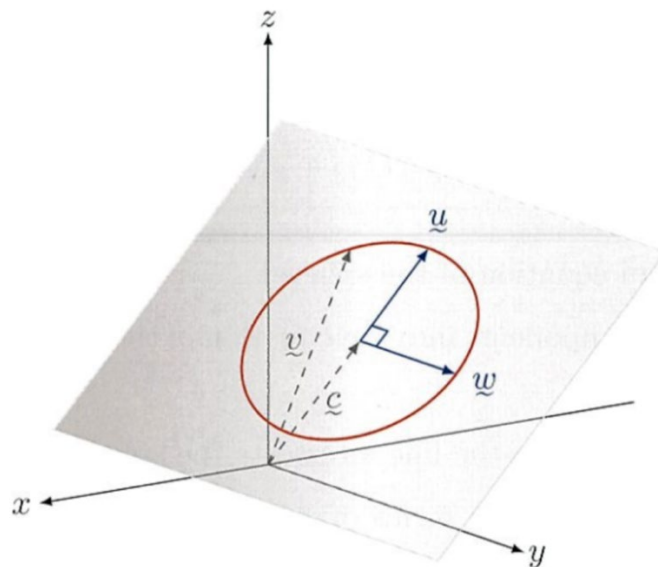
- (b) The straight line L has vector equation, 4

$$\vec{r} = \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \quad \lambda \in [-2, 5].$$

The point A has position vector $\mathbf{i} + \mathbf{j} - \mathbf{k}$.

By finding the reflection of A on the line L , calculate the area of the kite thus formed.

- (c) Consider a circle with radius r on some plane P containing two perpendicular vectors $\vec{u}$ and $\vec{w}$ both with length r .



Let the centre of the circle have position vector $\vec{c}$ and let $\vec{v}$ represent the position vector of any point on the circle.

- (i) Write the standard parametrisation for the circle centred at the origin with radius r in the xy plane. 1

- (ii) Use your previous answer to explain how 2

$$\vec{v} = \vec{c} + \vec{u} \cos \theta + \vec{w} \sin \theta$$

represents a circle in 3D space.

- (iii) Find the radius of the circle and verify that it is r . 1

- (d) Find the shortest distance between the lines: 3

$$l_1 = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} + \mu_1 \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad l_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \mu_2 \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$$

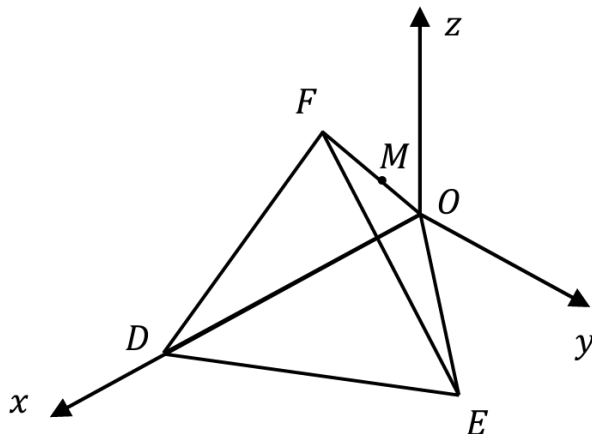
End of Question 2

Question 3 (12 marks)

- (a) Astronomers have traced the path of two asteroids travelling through space. At a particular time t , the position of asteroids Ceres and Pallas can be represented as shown below:

$$\begin{aligned}\text{Ceres: } & (-1 + t, 4 - t, -1 + 2t) \\ \text{Pallas: } & (-7 + 2t, -6 + 2t, -1 + t)\end{aligned}$$

- (i) Do the paths of the two asteroids cross? If so, where? 4
- (ii) Following these paths, will the asteroids collide? If so, where? 3
If not, find the closest distance between them.
- (b) The faces of tetrahedron $ODEF$ are comprised of equilateral triangles of side length 1 unit. Its base lies flat on the xy plane with vertices at O , $D(1,0,0)$ and $E\left(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right)$ as shown. Prove the coordinates of M , the midpoint of FO , is $\left(\frac{1}{4}, \frac{\sqrt{3}}{12}, \frac{\sqrt{6}}{6}\right)$. 3



- (c) Sketch the curve given by the equations: 2
- $$x = t^2, \quad y = \frac{1}{t^2}$$

End of Examination

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Mathematics Extension 2

Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 2. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

- Start Here** → 1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐

2023 Task 3

Question 3(c) solution

(d) (i)

3a	$\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{3}{2} \neq 1$ Hence, these numbers do not represent direction cosines of a line.		
3b	$AB = -2i + 3j - 4k$ and $\overrightarrow{AC} = -5i - 4j - 7k$ $\overrightarrow{BC} = \begin{pmatrix} -5 \\ -4 \\ -7 \end{pmatrix} - \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} -3 \\ -7 \\ -6 \end{pmatrix}$ $ \overrightarrow{AB} = \sqrt{2^2 + 3^2 + 4^2} = \sqrt{14}$ $ \overrightarrow{AC} = \sqrt{5^2 + 4^2 + 7^2} = \sqrt{90}$ $ \overrightarrow{BC} = \sqrt{3^2 + 7^2 + 6^2} = \sqrt{94}$ $\cos A = \frac{90 + 14 - 94}{2\sqrt{90}\sqrt{94}}$ $\angle BAC = \cos^{-1}\left(\frac{90 + 14 - 94}{2\sqrt{90}\sqrt{94}}\right)$ $= 81.9^\circ \text{ 1 d.p.}$	1 mark: determines $\overrightarrow{BC}$ 1 mark: calculates the lengths of AB, BC, CA 1 mark: calculates the angle	
3 (c)	DISTANCE = MAGNITUDE (PYTHAGORAS) $(-7 - -2)^2 + (k - 5)^2 + (3 - -7)^2 = (5\sqrt{14})^2$ $(-5)^2 + (k - 5)^2 + 10^2 = 350$ $(k - 5)^2 = 225$ $k - 5 = \sqrt{225} = \pm 15$ $k = \pm 15 + 5$ $k = 20 \quad k = -10$		
3d (i)	$\overrightarrow{OF} = \overrightarrow{OA} + \overrightarrow{AF} = \mathbf{a} + \mathbf{c}$ as $\overrightarrow{AF} = \overrightarrow{OC} = \mathbf{c}$ Thus $\overrightarrow{OF} = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} = 5i + 5k$	1 mark: correct calculations with $\mathbf{a}$ and $\mathbf{c}$ and correct reasoning 1 mark: correct vectors in terms of 5	

		and expresses in component form.	
3d (ii)	$\overrightarrow{FE}$ and $\overrightarrow{OB}$ are parallel and equal as the sides of the cube Hence, $\overrightarrow{FE} = \overrightarrow{OB} = \mathbf{b}$	. 1 mark: correct reason	
3d (iii)	$\overrightarrow{OE} = \overrightarrow{OF} + \overrightarrow{FE} = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} + \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ 5 \\ 5 \end{pmatrix} = \sqrt{5^2 + 5^2 + 5^2} = 5\sqrt{5}$	1 mark: gives the position vector of F 1 mark: correct answer with working	
3e	Let (a, b, c) be the perpendicular vector Then $(1, -3, 2) \cdot (a, b, c) = 0$ Also, $(2, 3, 1) \cdot (a, b, c) = 0$ $a - 3b + 2c = 0$ $2a + 3b + c = 0$ Adding, $3a + 3c = 0$, then $c = -a$ Also, $b = -\frac{a}{3}$ $\begin{pmatrix} a \\ a \\ -\frac{a}{3} \end{pmatrix} = \frac{a}{3} \begin{pmatrix} 3 \\ 3 \\ -1 \end{pmatrix}$ Thus, the vector perpendicular to l_1 and l_2 is $(3, -1, -3)$	1 mark: sets up dot product of direction vectors and the arbitrary vector (a, b, c) correctly 1 mark: correct evaluation and then the correct answer	

Question 4

4a (i)	$\overrightarrow{AB} = \begin{pmatrix} 1 \\ 13 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 8 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ -2 \end{pmatrix}$ $\tilde{r}_1 = \begin{pmatrix} 0 \\ 8 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \\ -2 \end{pmatrix} = \begin{pmatrix} \lambda \\ 8 + 5\lambda \\ 3 - 2\lambda \end{pmatrix}$	1 mark: Gives the correct equation of the straight line	
4a (ii)	$\tilde{r}_2 = \begin{pmatrix} 7 \\ 0 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 + 2\mu \\ -3\mu \\ 9 + \mu \end{pmatrix}$ Checking $\tilde{r}_1$ and $\tilde{r}_2$ for intersection $\begin{pmatrix} \lambda \\ 8 + 5\lambda \\ 3 - 2\lambda \end{pmatrix} = \begin{pmatrix} 7 + 2\mu \\ -3\mu \\ 9 + \mu \end{pmatrix}$ $\lambda = 7 + 2\mu$ $5\lambda + 8 = -3\mu$		

	$5(7 + 2\mu) + 8 = -3\mu$ $35 + 10\mu + 8 = -3\mu$ $13\mu = -43$ $\mu = -\frac{43}{13}$ Then, $\lambda = \frac{5}{13}$ Check: $3 - 2\left(\frac{5}{13}\right) = -\frac{43}{13} + 9$ $\frac{29}{13} \neq \frac{74}{13}$ $\therefore$ the lines do not intersect.	1 marks: Equates the l, j, k components and solves for λ and μ 1 mark: Tests the solution and gives the correct judgement	
4a (iii)	$\vec{AB} \cdot \vec{BC} = 0$ $= \begin{pmatrix} 1 \\ 5 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} x-1 \\ y-13 \\ z-1 \end{pmatrix} = 0$ $x-1 + 5y-65-2z+2 = 0$ $x+5y-2z+64 = 0$ Sub. x, y, z $7 + 2\mu - 15\mu - 18 - 2\mu + 64 = 0$ $-15\mu = 75$ $\mu = -5$ Then $C(-3, 15, 4)$	2 marks: correct answer from correct working 1 mark: Sets up $\vec{AB} \cdot \vec{BC} = 0$ and some minor error in working (must use the direction vector)	
4b	$\lambda = -2, \quad C \begin{pmatrix} 1-2 \\ -2-2 \\ 5+2 \end{pmatrix} = \begin{pmatrix} -1 \\ -4 \\ 7 \end{pmatrix}$ $\lambda = 5, \quad D \begin{pmatrix} 1+5 \\ -2+5 \\ 5-5 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \\ 0 \end{pmatrix}$	1 mark: determines the end points C and D and calculates length. 2 marks: finds the projection of CA onto L, or otherwise,	

	$CD = \sqrt{(6+1)^2 + (3+4)^2 + (7)^2} = 7\sqrt{3}$ Let $P(x, y, z)$ Project $\vec{CA}$ onto line L $\vec{CA} = \begin{pmatrix} 1-1 \\ 1-4 \\ -1-7 \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \\ -8 \end{pmatrix}$ $Proj_L \vec{CA} = \frac{\begin{pmatrix} 2 \\ 5 \\ -8 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}}{\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ $= \frac{2+5+8}{3} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = (5, 5, -5)$ Thus, the position vector of P is $(-1, -4, 7) + (5, 5, -5) = (4, 1, 2)$ $\begin{array}{ccc} A & \xrightarrow{+3} & P \\ \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} & \xrightarrow{+0} & \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} \\ & \xrightarrow{+3} & \end{array}$ $AB = \sqrt{(7-1)^2 + 0 + (5+1)^2} = 6\sqrt{2}$ $\text{Area of the kite} = \frac{1}{2} \times 7\sqrt{3} \times 6\sqrt{2} = 21\sqrt{6} \text{ sq. units}$	determines the foot of the perpendicular P. 1 mark: Uses a valid method to find P with some error 1 mark: finds the reflected point B 1 mark: calculates the area.	
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Question 5

5a (i)	<i>Ceres:</i> $(-1+t, 4-t, -1+2t)$ <i>Pallas:</i> $(-7+2t, -6+2t, -1+t)$ Consider the lines: $(-1+\lambda, 4-\lambda, -1+2\lambda)$ $(-7+2\mu, -6+2\mu, -1+\mu)$ For some values it $t = \lambda$ and μ Then, the equations of lines can be written as $(-1, 4, -1) + \lambda(1, -2, 2)$ and $(-7, -6, -1) = \mu(2, 2, 1)$ The direction vectors are not parallel as $(1, -2, 2) \neq k(2, 2, 1), k \in \mathbb{R}$	1 mark: tests for parallel	
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	Also, $d_1 \cdot d_2 =$ $1 \times 2 + -1 \times 2 + 2 \times 1 = 2 \neq 0$ Hence the lines are not perpendicular to each other. Checking for point of intersection: $\begin{pmatrix} -1 + \lambda \\ 4 - \lambda \\ -1 + 2\lambda \end{pmatrix} = \begin{pmatrix} -7 + 2\mu \\ -6 + 2\mu \\ -1 + \mu \end{pmatrix}$ $\begin{aligned} 2\mu - \lambda &= 6 \\ 2\mu + \lambda &= 10 \end{aligned}$ $4\mu = 14 \rightarrow \mu = 4$ Then, $\lambda = 2\mu - 6 = 2$ Test: $-1 + 2\lambda = -1 + \mu$ $-1 + 2(2) = -1 + 4$ $3 = 3 \text{ True}$ Hence, the paths do cross each other. Point of intersection $\begin{pmatrix} -1 + 2 \\ 4 - 2 \\ -1 + 2(2) \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$	1 mark: tests for perpendicular 1 mark: proves the lines do intersect 1 mark: finds the point of intersection	
5a (ii)	Using the result from (i), the asteroids reach the point (1,2,3) at $t = 2$ and $t = 3$ respectively. That is, they do not collide with each other. At time t seconds, let C be the position vector of Ceras and P be that of Pallas. Then the vector $\begin{aligned} \overrightarrow{CP} &= \begin{pmatrix} -1 + t \\ 4 - t \\ -1 + 2t \end{pmatrix} - \begin{pmatrix} -7 + 2t \\ -6 + 2t \\ -1 + t \end{pmatrix} \\ &= \begin{pmatrix} -6 + t \\ -10 + 3t \\ -t \end{pmatrix} \end{aligned}$ $ \overrightarrow{CP} ^2 = (t - 6)^2 + (3t - 10)^2 + t^2$ $= t^2 - 12t + 36 + 9t^2 - 60t + 100 + t^2$ Let $f(t) = 11t^2 - 72t + 136$ Minimum value of $f(t)$ when $t = \frac{72}{2 \times 11} = \frac{36}{11}$ Substituting, the minimum distance is 4.26 units.		

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